Name :NIKASH K R

Reg No : 411625243035

Department : B.TECH(AIDS)

Subject : Problem Solving Using Python

Batch : 03

PROJECT: Dice Game – 4 Players vs Computer

AIM

To develop a Python program that simulates a dice game between four players and the computer, where each rolls two dice per round and the highest total wins.

ALGORITHM

1. Start the program
2. Import required modules (random, time)
3. Create a function to roll two dice (1–6)
4. Ask number of rounds
5. Initialize scores for:
    - Player1, Player2, Player3, Player4
    - Computer
6. Repeat for each round:
    - Wait for Enter
    - Show rolling animation
    - Roll dice for all players + computer
    - Display results
    - Find highest score
    - Assign winner of round
    - Update scores

7. After all rounds:
    - Display final scores
    - Declare overall winner
8. Ask to play again
9. End

CODE

```
import random
import time

def roll_dice():
    return random.randint(1, 6), random.randint(1, 6)

def print_roll(name, roll):
    print(f"{name} rolled: {roll[0]} + {roll[1]} = {sum(roll)}")

def play_dice():
    scores = {
        "Player1": 0,
        "Player2": 0,
        "Player3": 0,
        "Player4": 0,
        "Computer": 0
    }

    rounds = int(input("Enter number of rounds: "))

    for i in range(1, rounds + 1):
        input(f"\n🎲 Press Enter to roll (Round {i})...")

        print("Rolling", end="")
        for _ in range(3):
            print(".", end="", flush=True)
            time.sleep(0.5)
        print("\n")

        rolls = {}
```

```
        # Roll for all players
        for player in scores:
            roll = roll_dice()
            rolls[player] = sum(roll)
            print_roll(player, roll)

        # Find winner of round
        max_score = max(rolls.values())
        winners = [p for p in rolls if rolls[p] == max_score]

        if len(winners) == 1:
            winner = winners[0]
            print(f"\n🏆 {winner} wins this round!")
            scores[winner] += 1
        else:
            print("\n🤝 It's a tie between:", ", ".join(winners))


        print("\n📊 Current Scores:")
        for p in scores:
            print(f"{p}: {scores[p]}")


    print("\n🏁 FINAL SCORE")
    for p in scores:
        print(f"{p}: {scores[p]}")

    max_final = max(scores.values())
    final_winners = [p for p in scores if scores[p] == max_final]

    if len(final_winners) == 1:
        print(f"\n🏆 {final_winners[0]} WON the game!")
    else:
        print("\n🤝 Game is a DRAW between:", ", ".join(final_winners))

def main_dice():
    while True:
        play_dice()
        again = input("\nPlay again? (yes/no): ").lower()
```

```
    if again != "yes":
        print("🙏 Thanks for playing Dice Game!")
        break

main_dice()
```

SAMPLE OUTPUT

```
Enter number of rounds: 2

🎲 Press Enter to roll (Round 1)...
Rolling...

Player1 rolled: 3 + 4 = 7
Player2 rolled: 6 + 2 = 8
Player3 rolled: 1 + 5 = 6
Player4 rolled: 4 + 4 = 8
Computer rolled: 2 + 3 = 5

🤝 It's a tie between: Player2, Player4

📊 Current Scores:
Player1: 0
Player2: 0
Player3: 0
Player4: 0
Computer: 0
```

RESULT

The Dice Game was successfully implemented for multiple players using Python. The program supports 4 players and a computer, handles multiple rounds, and determines winners based on dice outcomes.